Control Single Servo

I. API Introduction

```
The corresponding API for controlling a single bus servo is:
```

Arm_serial_servo_write(id, angle, time)

Function: Control the angle to which the bus servo should run.

Parameter explanation:

```
id: The ID number of the servo to control, range is 1~6. Each ID number
represents a servo, with the bottom-most servo having ID 1, increasing upward,
and the top-most servo having ID 6.

angle: The angle to which the servo should run. Except for servo 5 (ID=5), all
other servos have a control range of 0~180. Servo 5 has a control range of
0~270.

time: The time for the servo to run. Within the effective range, for the same
angle of rotation, the smaller the input time, the faster the servo movement.
Input 0 for the servo to run at maximum speed.

Return value: None.
```

II. Start Docker

After entering the main board desktop, open a terminal and run the following command to start the corresponding Dofbot container:

```
./Docker_Ros.sh
```

Then enter the following command in the Docker terminal to start jupyter-lab,

```
cd
jupyter-lab --allow-root
```

Then enter the main board's IP address followed by :9999 in the browser,

```
IP:9999         // Example: 192.168.1.11:9999
```

III. Code Content

Code path: /root/Dofbot/3.ctrl_Arm/3.ctrl_servo.ipynb

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device
# Create robotic arm object
Arm = Arm_Device()
time.sleep(.1)
```

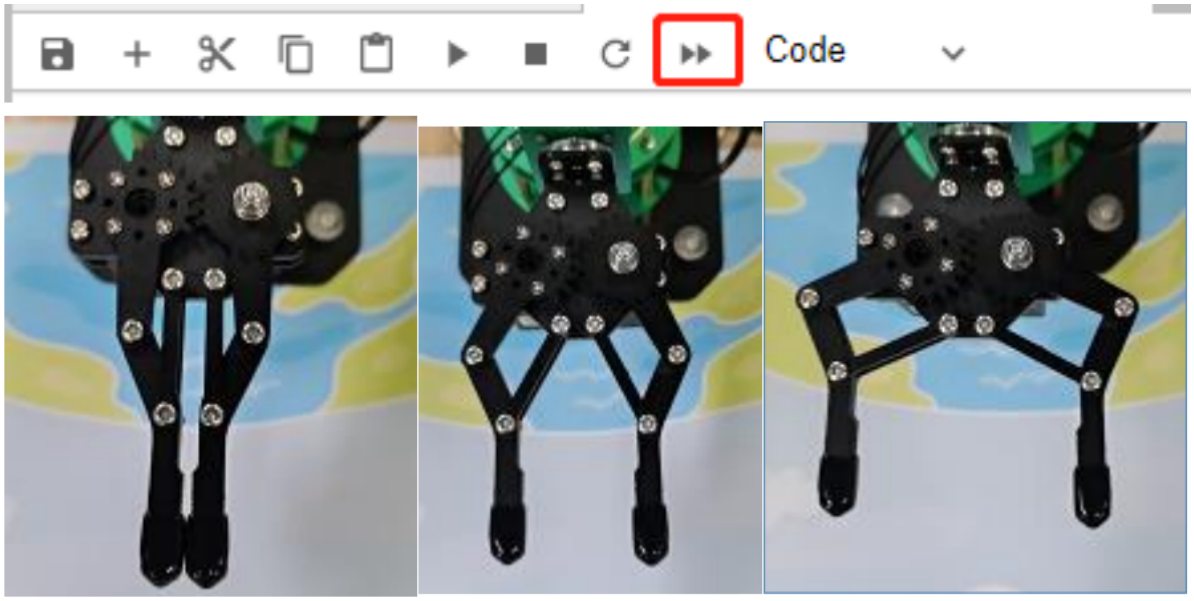
```
# Individually control a servo to move to a specific angle
id = 6
Arm.Arm_serial_servo_write(id, 90, 500)
time.sleep(1)
```

```
# Control a servo to cycle through angles
id = 6
def main():
 while True:
 Arm.Arm_serial_servo_write(id, 120, 500)
 time.sleep(1)
 Arm.Arm_serial_servo_write(id, 50, 500)
 time.sleep(1)
 Arm.Arm_serial_servo_write(id, 120, 500)
 time.sleep(1)
 Arm.Arm_serial_servo_write(id, 180, 500)
 time.sleep(1)

try :
 main()
except KeyboardInterrupt:
 print(" Program closed! ")
 pass
```

```
del Arm # Release the Arm object
```

Open the program file from jupyter lab, and click the "run entire notebook" button on the jupyter lab toolbar, you will see the robotic arm's gripper continuously changing angles.



To exit, click the stop button on the toolbar.

Code